

Should You Wear Earables When You Sleep?

Mobile Ubiquitous Computing CS7470

Replication Study Report

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ABSTRACT

As ear-worn wearable devices, also known as earables, gain popularity for applications like sleep tracking and health monitoring, user comfort during prolonged use, particularly during sleep, remains underexplored. This study replicates and builds on prior research evaluating the comfort and wearability of various earable designs during sleep. Due to logistical constraints, our replication study involved 6 earable models and 11 participants, each testing multiple devices over 2 nights. Using standardized comfort rating scales and questionnaires, we gathered both quantitative and qualitative feedback to assess fit, intrusiveness, and overall sleep experience. While some findings aligned with the original study, notable differences emerged in device rankings and user preferences, potentially influenced by variations in experimental design and user demographics. Our results highlight the complexity of designing earables that integrate seamlessly into sleep environments and reinforce the need for user-centered approaches in wearable technology development.

1. INTRODUCTION

As wearable technology continues to miniaturize and integrate into our daily lives, earables, wearable devices situated in or around the ear, have emerged as a platform for ubiquitous computing. These devices have been explored for applications like monitoring, activity recognition, interaction, etc. yet, despite their ubiquitous presence, one challenge has remained under-explored: user comfort on extended periods of passive wear during sleep.

Our project investigates the ability to comfortably wear such devices overnight which may help in extended passive applications like sleep monitoring and tracking studies. However, comfort is both subjective and multi-dimensional, influenced by factors such as ear anatomy, device form factor, materials and sleep posture. Through a combination of prior-research review and user feedback we aim to contribute to the growing research around human-centered design of wearable devices. As such, our project analyses the question, “how can we ensure that these devices integrate into the unconscious moments of our daily lives without causing discomfort or disruption?”

2. BACKGROUND

Earables, wearable devices designed to be worn in or around the ear, have garnered significant attention for their potential in continuous health monitoring, sleep tracking, and user interaction. While earlier research emphasized wearable sensing capabilities,

recent studies have begun to address the critical aspect of user comfort, especially during sleep.

User comfort remains a primary factor in the adoption of sleep-related wearables. A study evaluating a digital wellness device designed to mask environmental noise found that participants experienced a 32.5% reduction in perceived sleep onset latency. Importantly, 86% reported falling asleep more easily, and 82% felt their overall sleep quality improved, underscoring the significance of comfort and perceived effectiveness in user experience.[3] A randomized crossover trial found that using a wearable sleep tracker improved perceived sleep quality in healthy participants, suggesting that using such devices can positively influence sleep behaviors.[4]

The choice of materials in earable design also significantly impacts comfort during sleep. Viscoelastic materials, for instance, have been employed in in-ear EEG sensors to provide a stable interface with the skin by expanding and redistributing pressure evenly along the contact surface. This design reduces mechanical artifacts caused by motion or cardiac pulsation, which is particularly beneficial during sleep when movement is minimal.[5] In contrast, silicone earbuds, while commonly used, may offer suboptimal conformance to the ear canal's shape, potentially leading to discomfort or instability during sleep.[5] Therefore, selecting materials that adapt to individual ear anatomies without exerting excessive pressure is essential for prolonged wearability.

User feedback also plays a vital role in assessing the comfort of earables. In a study involving a dry-contact ear-EEG device, participants reported that the ear plugs were easy or very easy to use, with comfort levels improving over successive nights. This suggests that users may adapt to the device over time, highlighting the importance of an initial acclimatization period.[6] Similarly, consumer-grade sleep buds like the Anker SoundCore Sleep A20 have been developed with ergonomics in mind, featuring passive noise-canceling technology and designs tailored for side sleepers. These devices aim to provide a comfortable and secure fit throughout the night, addressing common discomfort issues associated with traditional earbuds.[7] Furthermore, the design of earables should accommodate various sleep positions. Devices that maintain stability without causing discomfort when lying on the side or back are more likely to be accepted by users. Adjustable components that allow users to customize fit can also enhance comfort and reduce the likelihood of the device dislodging during sleep.

Beyond physical design, integrating features that promote relaxation can enhance the overall sleep experience. For example, the 'Cove' wearable applies gentle vibrations behind the ears to

reduce stress and improve sleep quality. Such non-invasive stimulation methods can complement the physical comfort of the device, contributing to a more restful sleep.[8] In conclusion, the wearability and comfort of earables during sleep are influenced by material choice, ergonomic design, user adaptability, and the integration of comfort-enhancing features. Our replication study aims to build upon these insights by developing earables that prioritize user comfort, thereby facilitating better sleep quality and user adoption.

3. EXPERIMENT REPLICATION

The original study “Wearability and Comfort of Earables During Sleep” [2] and our study focuses on the same concepts of determining the comfortability and wearability of the earable devices and ranking them based on the participants responses. However, due to multiple restrictions on earable obtainability, timelines, and distance between team members, the study design of our replication experiment is slightly different than the original.

3.1 Original Experiment

The original study revolved around 7 different earables that participants would be sleeping with. The original 7 included earable models of i12, eSense, HolyHigh, Cosinuss, SBH82D4, Z8, and K8. These 7 models represented a broad range of earable designs and varying materials, including rigid or soft materials, hook designs, and inner ear design.



Figure 1: Earables in original study

For the study, 14 individuals were recruited and given in-person instructions about the study. Each participant would then receive all 7 types of earables along with their questionnaires. The study was conducted in a naturalistic setting, where participants used the devices at home during sleep. Every night the participant would test a different earable and wear the earable to sleep. Each earable is only worn once and the next morning the participant would complete the questionnaire for his or her designated earable worn. The survey included:

- The CRS (Comfort Rating Scales)
- Questions assessing the device’s influence on sleep and overall suitability
- Whether the participant kept the device in all night and how many times it fell out
- General concerns and demographic information
- Open-ended feedback

Once all devices were tested, participants were asked to rank them in order of preference based on their sleep experience.

3.2 Our Experiment

The procedures for our experiment revolved around 6 different earables, instead of 7, that participants would be sleeping with. Since there was great difficulty obtaining Cosinuss and SBH82D4

from the original study, our team kept 5 of the original earables: i12, eSense, HolyHigh, K8, and Z8. In addition, we added in Skullcandy Jib+ into the number of earables to make 6 designs in total. To split up obtaining the earables evenly, each team member obtained 2 different types of earables. Since there are only 6 different models, eSense and i12 by two members.



Figure 2: Earables in replication study

The procedures of the replication experiment was kept as closely the same to the original as possible to ensure as little deviation from the findings of the replication to the original. For the experiment, our team was able to gather 11 individuals in total. Due to geographic separation of team members, we were unable to implement the Williams design fully. Each member was tasked with gathering 2-3 members for the study. However, due to the restriction of our teammates in different locations, the process of splitting up the earables required the change in each participant being able to wear each earable. Instead, each participant would wear one earable for two nights. After two nights, he or she would switch to a different earable and wear that device for another two nights. This adaptation ensured participants had enough time to assess comfort while enabling us to collect comparative data across devices. Like the original study, each person was asked to complete the questionnaire when waking up in the morning. Following the principle of adhering as closely to the original study as possible, the questionnaire that were given to the participants in the replication study is identical to the questions that were given to the participants in the previous study.

3.3 Differences/Challenges

Our study was shaped by several key challenges, many of which were anticipated in our original proposal:

- Device availability: We were unable to source two of the original devices (Cosinuss, SBH82D4) despite extensive searching. To address this, we substituted with a comparable market available device (Skullcandy Jib+) that had a new design that would not conflict with the original 5 designs we already obtained and documented all device-related differences, preserving the study’s validity.
- Participant distribution and logistics: With team members located in different regions, we could not provide all devices to all participants or conduct in person procedures. Instead, each participant tested a subset of devices, and all instructions and questionnaires were standardized to be the same for all individuals. While this reduced the number of overall survey responses in half, this new procedure solidified the collected data on each type of earable since the participants wore the devices more than once.
- Consistency and data collection: Despite changes, we upheld consistency in data collection by using the same questionnaire and instructing participants to complete it in the same context (immediately upon waking, reflecting sleep experience).

3.4 Comparison

Despite the modifications to our replication study's structure, procedures, and logistics, the core goal remained unchanged: to evaluate the comfortability and wearability of earables during sleep. Every adaptation we made was grounded in preserving the integrity of this objective while working within practical constraints.

First, we maintained a representative selection of earable designs and materials by including five models from the original study and substituting the unavailable ones with a comparable market-available device. This ensured we could still explore a diverse design space in terms of fit, attachment method, and physical properties—critical factors in comfort and wearability. Second, although each participant did not test every single earable (as in the original study), they still tested multiple devices across multiple nights, providing valuable comparative data. In fact, extending the trial period to two nights per device allowed participants to more thoroughly assess comfort over time, potentially increasing the reliability of their responses. Third, the identical questionnaire preserved consistency in how comfort and wearability were measured. This means our data remains directly comparable to the original study, and participant feedback continues to reflect the same core dimensions of interest: fit, intrusiveness, device retention, and sleep interference. Lastly, even with remote coordination and varying participant locations, we preserved the naturalistic, in-home sleep environment from the original study. This allowed us to capture real-world usage scenarios, which is essential to understanding how wearable technology performs in everyday life.

In summary, although our study involved logistical and procedural changes, each decision was intentionally made to support the original study's goal of understanding how comfortable and wearable earables are during sleep. Our methods were adapted—but not compromised—in our effort to produce meaningful, valid, and applicable findings.

4. RESULTS

Our experiment included 11 participants, 3 females and 8 males, with an average age of 31. These participants were tasked with submitting a questionnaire each morning that they wore an earable. Through these results, we were able to gain insight into valuable metrics and statistics based on tests used during the original study. See Appendice 1: Raw Survey Results, to view the data used in our analysis as well as the questions asked in the survey.

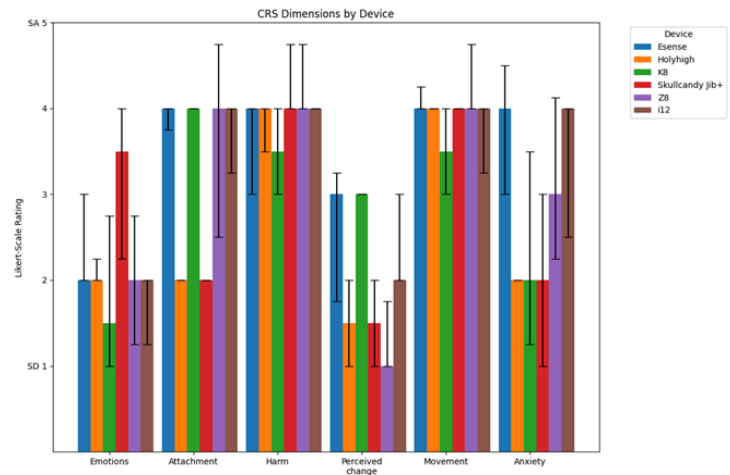
4.1 Quantitative Results

The Comfort Rating Scale (CRS), instructions for measuring the comfort of wearable computers [1], is the primary source used as a guide in the original studies questionnaire. In accordance with the CRS, these questions from the survey (asked on a Likert-Scale) align to the following CRS dimensions:

- **Emotions:** "I am worried about how I look when I wear this device"
- **Attachment:** "The device fits well or is easy to adapt", "attaching the device is easy to learn"
- **Harm:** "The device reduces my sleep quality"
- **Perceived change:** "I could imagine wearing the device regularly during sleep"

- **Movement:** "The device affects my sleep position"
- **Anxiety:** "I am concerned about having Bluetooth close to my head", "I am concerned about having batteries close to my head"

Figure 3 – CRS Dimensions by Device



In Figure 3 each of the CRS dimensions across the devices, the median scores and interquartile ranges are displayed. From this data we are able to make unique observations:

- For most devices, participants disagreed that they would be worried about how they looked wearing the device.
- Attachment varied significantly between devices with K8 considered to be the easiest to attach.
- Harm was most prevalent in Z8 and the Skullcandy Jib+ devices but each device was considered harmful.
- The Perceived Change of the devices also varied, but none of the devices were deemed as likely to be worn during sleep aside from some responses where Esense was the device.
- Most devices affected movement but K8 was considered the least invasive.
- Participants felt that the Esense caused the most concern about Bluetooth and batteries near their heads while the opposite was true for the Skullcandy Jib+.

Scores for Emotions differed from the original studies results. In the original study, participants were not concerned about the way they looked while sleeping. Whereas in our study, participants leaned more towards a minimal amount of concern. Another dimension with similar results, Anxiety yielded much more concern in our study compared to the original. Another difference in our study would be the way our participants perceived Attachment for the Holyhigh device. Participants found the Holyhigh earable to be more difficult to attach and less comfortable to wear. Overall, our results varied significantly from the original studies.

Device	Rank	Score	Fell Off	Taken Out
Esense	1	5	22	5
K8	2	3	0	0
i12 (Airpods)	3	3	23	4
Holyhigh	4	0	2	1
Skullcandy Jib+	5	0	10	1
Z8	6	0	7	6

Table 1: Preference Ranking and count of times device was removed

Table 1 displays the summed responses to the questions for the times the device fell out and the times the device was taken out from the questionnaire. Scores are based on responses from the participants ranking each device. If the device was listed first, it was given a value of 1 while the second device was given a value of 0. It's important to note that this question was only asked on the fourth night. Given that only three devices were given values and that both the i12 and K8 had the same preference scores, the values for the times device were taken/fell out were used to place K8 above i12. The same sort of ordering was used to determine the rank of the rest of the devices.

The data within Table 1 differs somewhat from the original study. In the ordering of the original, participants ranked the Esense device as one of the least preferred in the 6th place, followed by the Holyhigh in 7th place, whereas in our study, the Esense was listed 1st and the Holyhigh 4th. These results could be explained by differences in available devices used in our study. The rankings could also be due to only having each participant wear two devices. It's likely the latter causation behind the difference in results because while the original had more devices to be ranked, we had fewer to gather preference scores for.

4.2 Qualitative Results

The qualitative questions we asked participants were: "Did you have any general concerns?" and "If you had any other thoughts or feedback, feel free to add those here". These questions were optional and the K8 was the only device with no input from participants. The following list highlights the key insights and general trends per device.

- **Esense**
 - Impact on hearing ability
 - Discomfort caused by device while laying on side
 - General discomfort
- **Holyhigh**
 - Tight fit
 - Pain/discomfort when sleeping
- **I12**

- Mixed responses varying between comfortable/uncomfortable
- Fear of losing the device
- Impact on hearing ability
- Loose fit

- **Skullcandy Jib+**

- Fear of losing device

- **Z8**

- General discomfort

5. CONCLUSION/REFLECTION

Our aim of this replication study was designed to investigate the comfort and wearability of ear-worn devices during sleep. It was hoped that the results of the study would be able to replicate and build on the findings of the original ISWC 2021 study. Our team was able to collect meaningful quantitative and qualitative data involving six different headset designs and 11 participants through a modified but methodologically consistent replication study.

5.1 Positive Aspects

Even with the challenges of procurement and distribution, our team kept the core structure of the original experiment. The survey instrument let us make consistent comparisons, and our adapted participant rotation—wearing two sets of equipment over four nights—let us get more insight into long-term comfort. The participants were engaged throughout the study, and the results of the CRS (Comfort Rating Scale) were meaningful. They confirmed some of the initial findings and showed some differences from the initial findings. The Esense headset had the highest comfort and preference scores in our replication study, but lower scores in the original study. This shows how important the user's environment and test conditions are.

5.2 Discrepancy Between Our Intended Outcome and Actual Result

We encountered a number of challenges. The most significant impact of this endeavor was that the endeavor was unable to fully replicate the original, obtaining only five of the seven devices and introducing the Skullcandy Jib+ as a replacement. Furthermore, given that the participants had the opportunity to trial only two of the seven devices (as opposed to the full complement of seven in the original study), the comparative rankings may not have achieved the desired level of comprehensive analysis. This limitation can be attributed to the limited exposure experienced by the participants. Finally, disparities in demographics and lifestyle factors like sleep position, sensitivity to sound, etc. have the potential to result in variations in outcomes, which are challenging to regulate.

5.3 Were the Objectives of the Project Successfully Achieved?

Largely, yes. We accomplished our core objective: replicating and evaluating the comfort and suitability of headphones during sleep. While the limitations of our device and participants changed some of our conclusions, our methodology remained consistent. Our replication revealed new insights, including a higher concern for Bluetooth proximity among modern users and variability in perceived hazards across devices.

5.4 Broader Implications

Our findings suggest that the perception of comfort in headphones is influenced by individual differences and the context in which they are used. These findings also suggest that commercial headphone designs could benefit from taking different sleep positions and individual sensitivities into consideration, especially pressure points and battery locations. Additionally, while several headphones were designed with sleep in mind, the discomfort they cause might indicate a potential gap in user-centered ergonomics that could be addressed in future iterations of the hardware.

5.5 Connection Points to Future Work

Building on our replication study, future studies should aim for the followings:

- Investigate user adaptation over extended periods—two to three weeks should be sufficient.
- It is imperative to expand the participant pool in order to enhance the generalizability of the study's findings.
- Expand wider & more adaptive options like discover form-fitting or adaptive materials for a smarter custom fits.
- The inclusion of physiological sleep data (e.g., sleep stage tracking) alongside subjective comfort measures is essential.

Furthermore, the integration of user comfort studies with concurrent sensing functionality evaluations (e.g., heart rate or sleep apnea monitoring) would be in alignment with the prevailing trends in wearable research, as highlighted in the ISWC 25-year retrospective analysis. Our study provides a crucial piece of the puzzle, demonstrating that comfort is not just a bonus and it's essential for successful ubiquitous technology integration.

6. REFERENCES

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7. APPENDIX

7.1 Raw Survey Results

Click [here](#) to view results.

7.2 CRS Dimensions by Device

